3D Trajectory Simulation and Animation of a Low Earth Orbit (LEO) Satellite Using ODE45 Numerical Solver

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Problem Statement

The objective of this work is to develop a three-dimensional simulation and visualization of a Low Earth Orbit (LEO) satellite propagating around the Earth, as illustrated in the schematic

Figure 1. The satellite's orbital trajectory is numerically simulated in an inertial reference frame using the classical 2-body orbital dynamics. The simulation results are rigorously validated through the conservation of angular momentum and mechanical energy, ensuring physical consistency. Following validation, a high-fidelity animation of the satellite's orbital motion is generated, featuring a hyper-realistic Earth model to enhance physical intuition and visual interpretation of the orbital dynamics.

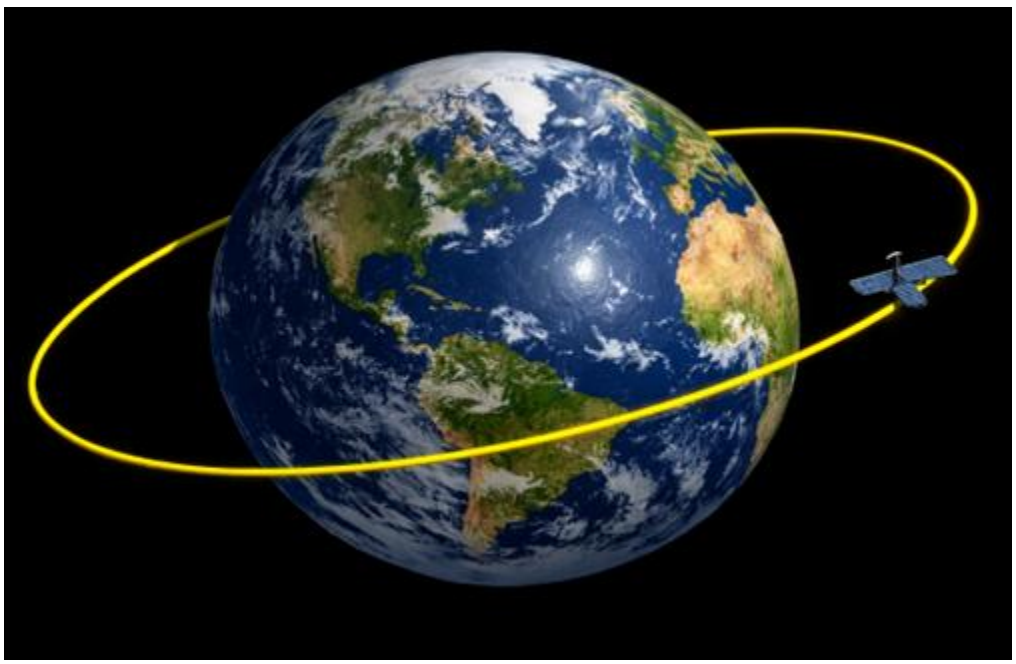

Figure 1: A Schematic Illustration of a Low Earth Orbit (LEO) Satellite Trajectory

To derive the orbital EoMs of a Low Earth Orbit (LEO) satellite in a 3D plane, an Earth-centered inertial coordinate system was considered with the origin at the Earth's Center of Mass (CoM). Let us assume the position of the satellite at any given time is at point P(*x, y, z*), which is defined by the position vector $\vec{\boldsymbol{r}}$.

Gravity Model:

Let the gravitational pull experienced by the satellite be $\overrightarrow{F_g}$, which can be resolved into three components F_{g_x}, F_{g_y} and F_{g_z}. According to Newton's Law of Gravitation, the magnitude of the gravitational force can be given as,

$$F_g = \frac{GMm}{r^2} \quad \text{----------} \quad \textbf{(1)}$$

where, G is the Gravitational Constant
M is the Mass of the Earth
m is the Mass of the Satellite
r is the Distance of the Satellite from the CoM

Rewriting equation (1), we get,

$$F_g = mg_0 \frac{R_E^2}{r^2} \quad \text{----------} \quad \textbf{(2)}$$

[∵, acceleration due to gravity on the Earth, $g_0 = \frac{GM}{R_E^2}$]

Now, we resolve $\overrightarrow{F_g}$ into its two perpendicular components as follows:

$$\overrightarrow{F_g} = F_{g_x}\hat{\imath} + F_{g_y}\hat{\jmath}$$

$$\Rightarrow \overrightarrow{F_g} = F_g \cos\theta\, \hat{\imath} + F_g \sin\theta\, \hat{\jmath}$$

$$\Rightarrow \overrightarrow{F_g} = -mg_0 \frac{R_E^2}{r^2} \cos\theta\, \hat{\imath} - mg_0 \frac{R_E^2}{r^2} \sin\theta\, \hat{\jmath} \quad \text{----------} \quad \textbf{(3)}$$

Newton's 2nd Law of Motion:

By combining Newton's 2nd Law of Motion with the Gravity Model, we get,

$$m\ddot{x} = F_{g_x} = -mg_0 \frac{R_E^2}{r^2} \cos\theta \quad \text{----------} \quad \textbf{(4)}$$

$$m\ddot{y} = F_{g_y} = -mg_0 \frac{R_E^2}{r^2} \sin\theta \quad \text{----------} \quad \textbf{(5)}$$

where, $\ddot{x}$ is the acceleration of the satellite along the X-axis, and $\ddot{y}$ is the acceleration along the Y-axis.

Rewriting the polar coordinates into the 3D Cartesian coordinates, we get,

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2+z^2}} \quad \text{and} \quad \sin\theta = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

Hence, equations (4) and (5) become,

$$\ddot{x} = -g_0 \frac{R_E^2\, x}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(6)}$$

$$\ddot{y} = -g_0 \frac{R_E^2\, y}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(7)}$$

Similarly, based on equations (6) and (7), the satellite acceleration along the Z-axis can be formulated as follows,

$$\ddot{z} = -g_0 \frac{R_E^2\, z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(8)}$$

Therefore, equations (6), (7), and (8) are the required EoMs for the LEO satellite orbiting around the Earth in a 3D coordinate system, which are coupled, nonlinear second-order ODEs. Solving these differential equations will yield the position and velocity of the LEO satellite around the Earth as a function of time.

Numerical Simulation of the EoMs with ODE45 Numerical Solver:

To solve this system of nonlinear equations, the second-order ODEs are reformulated as a system of first-order differential equations by assuming the three position components and their respective velocity components as the state variables of the system.

Let the state variables be,

$s_1 = x$	$s_4 = \dot{x}$
$s_2 = y$	$s_5 = \dot{y}$
$s_3 = z$	$s_6 = \dot{z}$

Hence, the system of first-order ODEs can be obtained as follows:

$$\dot{s}_1 = s_4 \quad \text{----------} \quad \textbf{(9)}$$

$$\dot{s}_2 = s_5 \quad \text{----------} \quad \textbf{(10)}$$

$$\dot{s}_3 = s_6 \quad \text{----------} \quad \textbf{(11)}$$

$$\dot{s}_4 = -g_0 \frac{R_E^2\, s_1}{({s_1}^2 + {s_2}^2 + {s_3}^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(12)}$$

$$\dot{s}_5 = -g_0 \frac{R_E^2\, s_2}{({s_1}^2 + {s_2}^2 + {s_3}^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(13)}$$

$$\dot{s}_6 = -g_0 \frac{R_E^2\, s_3}{({s_1}^2 + {s_2}^2 + {s_3}^2)^{\frac{3}{2}}} \quad \text{----------} \quad \textbf{(14)}$$

The initial conditions at $t_0 = 0$ sec are assumed as follows:

$s_1(0) = 3858.213$ km	$s_4(0) = -0.863$ km/s
$s_2(0) = -5798.143$ km	$s_5(0) = -0.542$ km/s
$s_3(0) = 14.693$ km	$s_6(0) = 7.497$ km/s

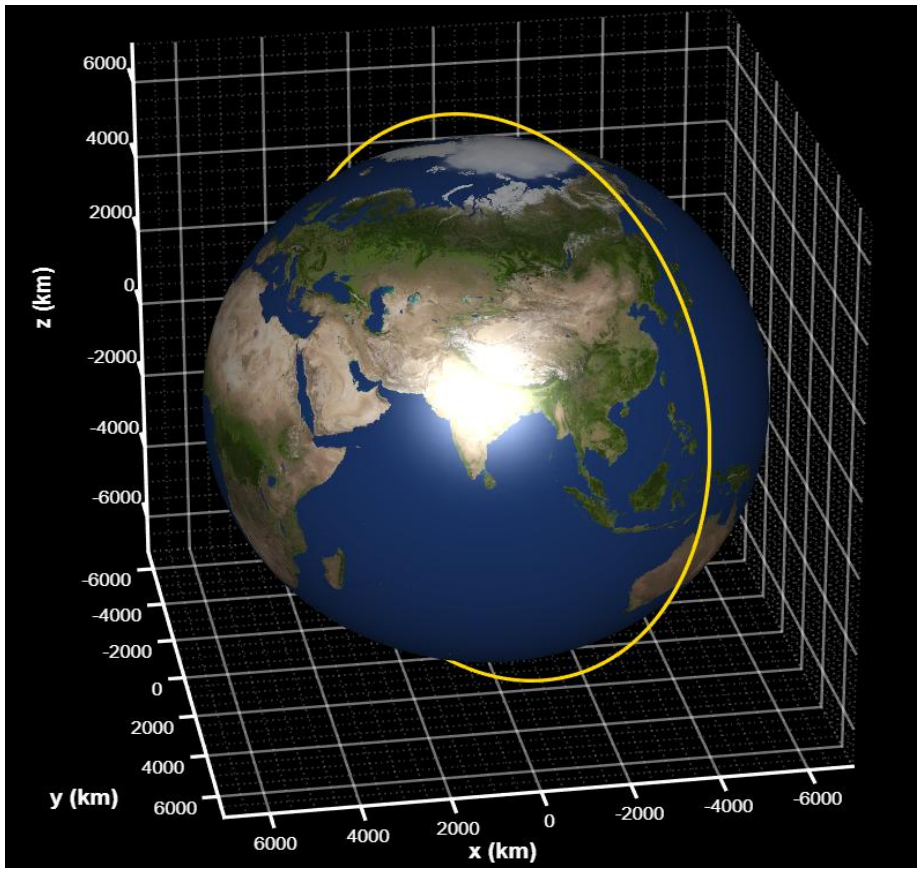

Figure 2: LEO Satellite Trajectory (The Yellow Ring) Simulation Around the Earth

Using the ODE45 solver in MATLAB, the LEO satellite's position was computed as a function of time, and its orbital plane was plotted around a hyper-realistic Earth model in 3D space, as evident in **Figure 2**. The 3D visualization of the orbital plane reveals that the satellite is rotating along a circular trajectory at a specific inclination.

From the simulation results, it was found that the final position of the satellite at 30,000 seconds was approximately 6950 km from the center of the Earth, and its velocity was 7.58 km/s.

Verification of the Simulation Results:

To verify the simulation results obtained from the ODE45 Solver, the total specific energy and angular momentum of the satellite were observed at every time step, as they remain conserved unless an external or non-conservative force is applied to the satellite.

We know that,

Total Specific Energy, $E = \frac{1}{2} V^2 - \frac{g_o R_E^2}{r}$ ---------- **(15)**

and Angular Momentum, $\vec{H} = \vec{r} \text{ x } \vec{v}$, ---------- **(16)**

where, $\vec{v}$ is the velocity vector of the satellite

As the specific energy and angular momentum of the satellite at each time step were computed, the results were compared with the initial specific energy and angular momentum to calculate the relative errors. Therefore, the relative error in the total energy at each time step can be given by,

$$\varepsilon = \frac{|E(t) - E(0)|}{|E(0)|}$$

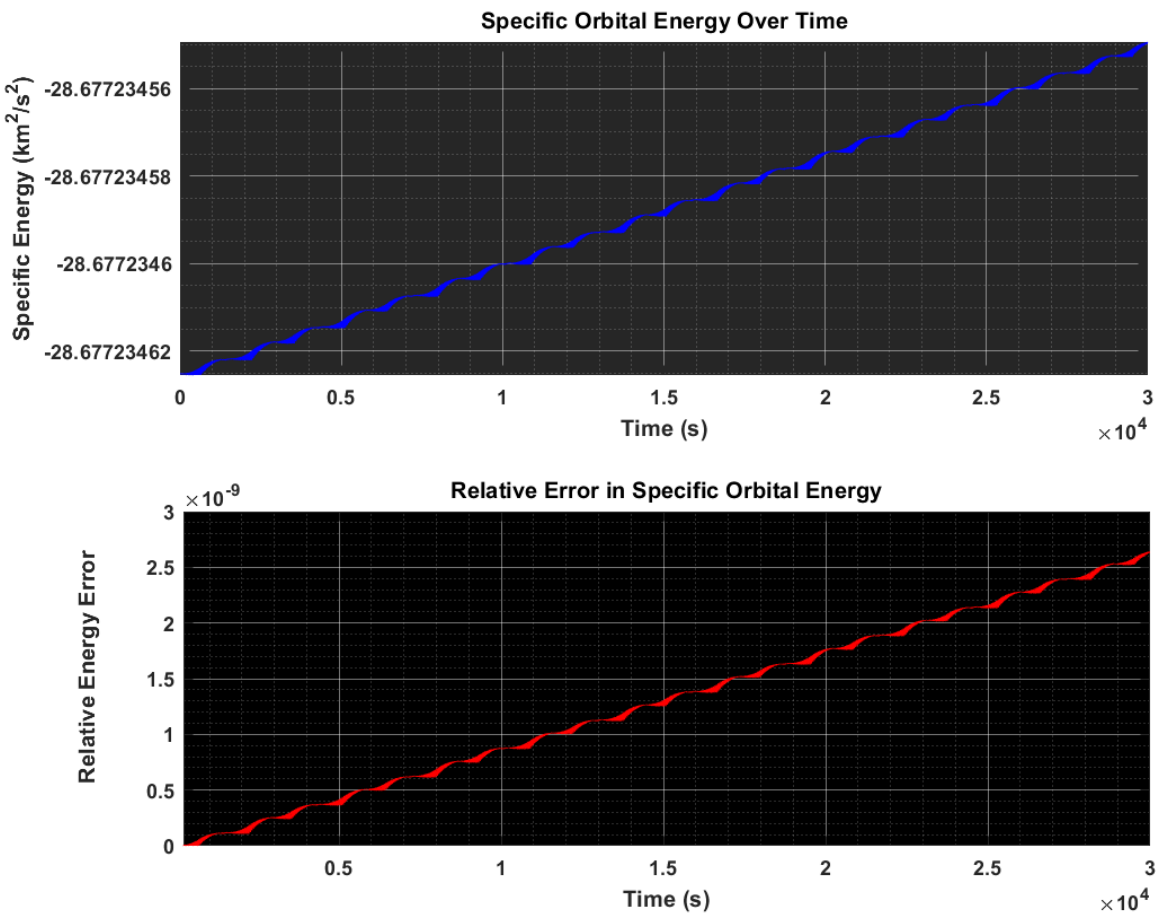

Figure 3: (i) Specific Orbital Energy Over Time; (ii) Relative Error in Specific Orbital Energy

As observed in **Figure 3(i)**, the specific orbital energy is constant for all time up to the eighth decimal place. However, small fluctuations still exist in the result due to the numerical integration errors. For most scientific applications, including precise orbital simulations, relative error < 10^{-6} is considered excellent. Meanwhile, in the plot of relative energy error for the LEO satellite, the error is in the range of 10^{-9}, which is illustrated in **Figure 3(ii)**.

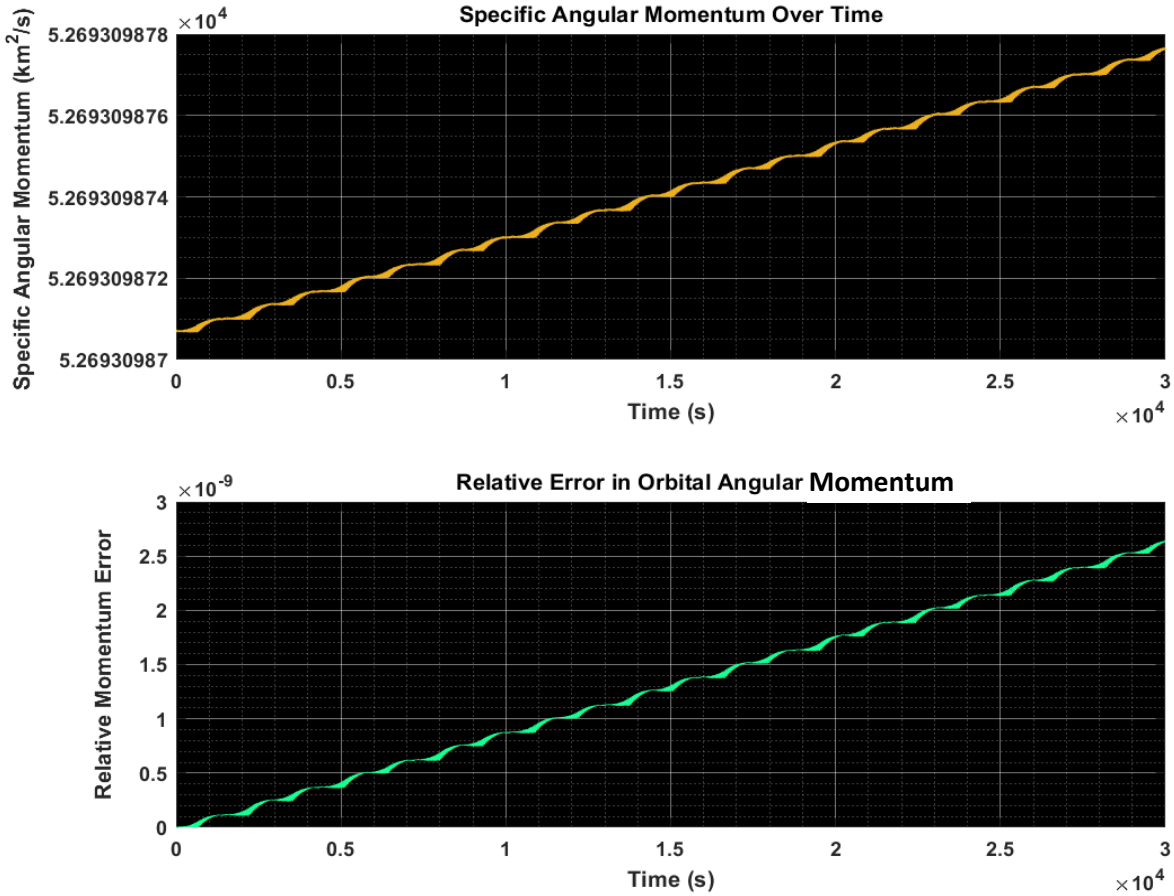

Figure 4: (i) Specific Angular Momentum Over Time; (ii) Relative Error in Orbital Momentum

Similarly, the satellite's specific angular momentum is constant for all time up to the eighth decimal place, as shown in **Figure 4(i)**, and its relative error is also in the range of 10^{-9}, which can be seen from **Figure 4(ii)**. These results validated the accuracy and precision of the acquired simulation results.

Animation of the LEO Satellite's Orbital Motion:

To complement the numerical simulation of the satellite's orbital trajectory acquired using the ODE45 solver, a three-dimensional visualization framework was developed to represent the inertial motion of the Low Earth Orbit (LEO) satellite around a realistic Earth model. The visualization operates purely as a post-processing step and does not influence the numerical integration of the equations of motion.

Earth Model and Reference Frame:

The Earth was modeled as a rigid sphere centered at the origin of an inertial Cartesian reference frame, as illustrated in **Figure 5**. The radius of the sphere corresponds to the mean Earth radius

used in the orbital dynamics formulation. A high-resolution NASA Blue Marble texture was mapped onto the spherical surface to enhance visual and provide geographic realism while preserving the underlying geometric fidelity. The Earth remained fixed throughout the animation, consistent with the inertial reference frame used for the orbital equations of motion.

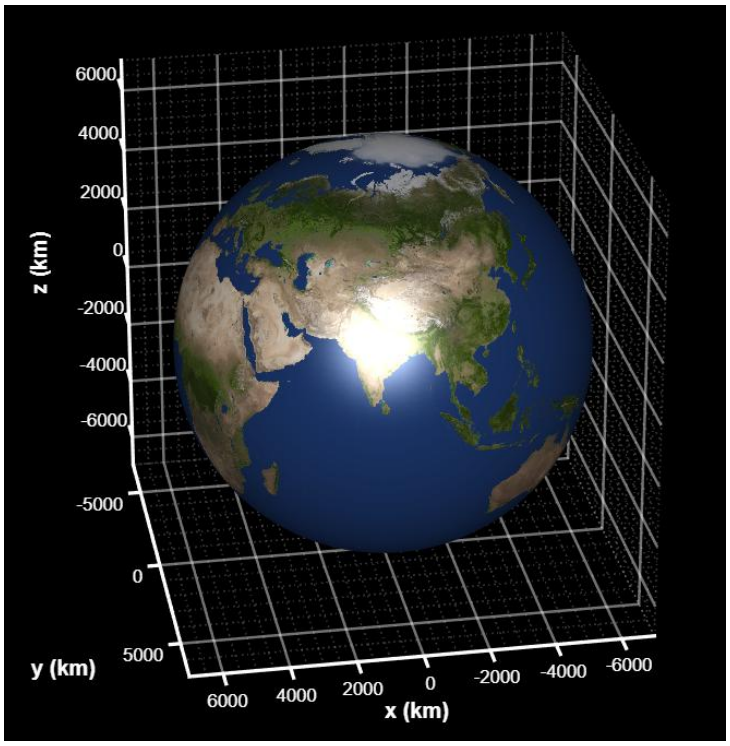

Figure 5: Visuals of a Hyper-Realistic Earth Model

<u>Satellite Geometry Modeling</u>:

The satellite was modeled as a rigid assembly composed of a central bus and two solar panels. The complete satellite geometry was defined in a local body-fixed coordinate system centered at the satellite's center of mass:

- The **satellite bus** was represented as a rectangular cuboid, defined by eight vertices and six planar faces.
- The **solar panels** were modeled as thin rectangular plates rigidly attached to opposite sides of the bus, symmetrically positioned along the lateral axis.

All geometric dimensions were specified in normalized units and subsequently multiplied by a visual scaling factor. This separation between physical dimensions and visualization scaling ensured that the satellite remained visible in the context of Earth-scale distances without affecting the numerical orbital solution. The satellite geometry is demonstrated in **Figure 6**.

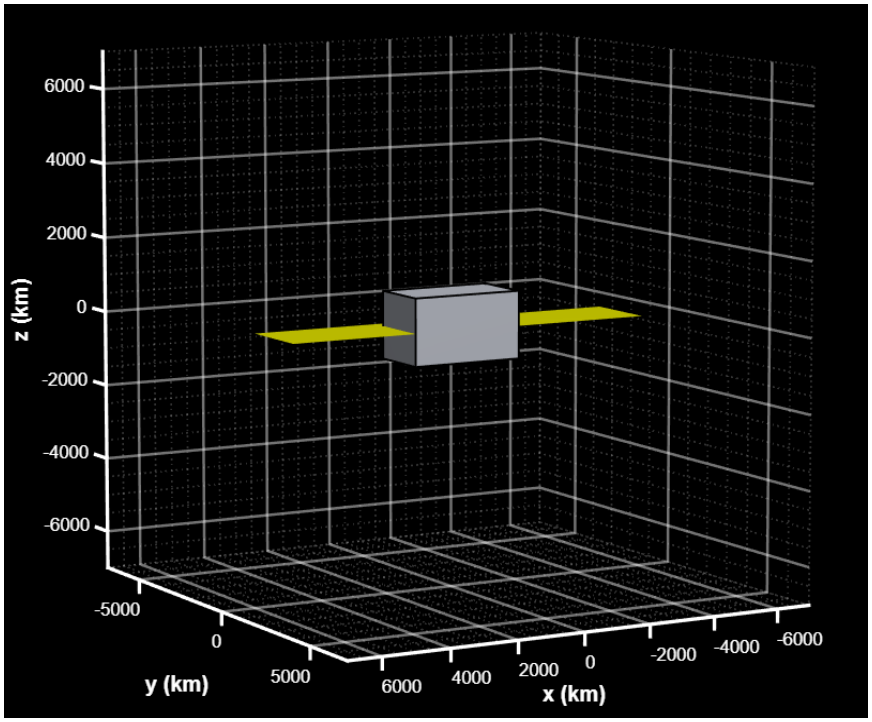

Figure 6: Visuals of a LEO satellite Geometry

<u>Coordinate Transformation and Frame-by-Frame Motion</u>:

The satellite's translational motion was driven directly by the numerical solution of the orbital equations of motion. At each animation frame, the satellite's inertial position vector was extracted from the ODE45 solution as

$$\mathbf{r}(t_k) = [x(t_k) \quad y(t_k) \quad z(t_k)]$$

The satellite geometry, initially defined in the local body-fixed frame, was translated into the inertial frame by adding this position vector to every vertex of the satellite bus and solar panels. The inertial position of each vertex at time t_k was therefore given by

$$\mathbf{v}_i^{\text{inertial}}(t_k) = s\,\mathbf{v}_i^{\text{body}} + \mathbf{r}(t_k),$$

where s denotes the visual scaling factor. In this work, the satellite orientation was held fixed, and only translational motion was animated. This assumption is consistent with the scope of the study, which focuses on orbital kinematics rather than attitude dynamics.

<u>Rendering Strategy</u>:

A perspective camera model and depth-aware rendering were employed to ensure physically consistent visual ordering between the Earth, satellite, and orbit, as elucidated in **Figure 7**. Lighting and shading were applied uniformly to the scene to preserve surface continuity and depth perception while avoiding visual artifacts near the poles of the Earth model. The animation was generated by sampling the numerical solution at discrete time intervals, allowing smooth visualization of the satellite motion while maintaining computational efficiency.

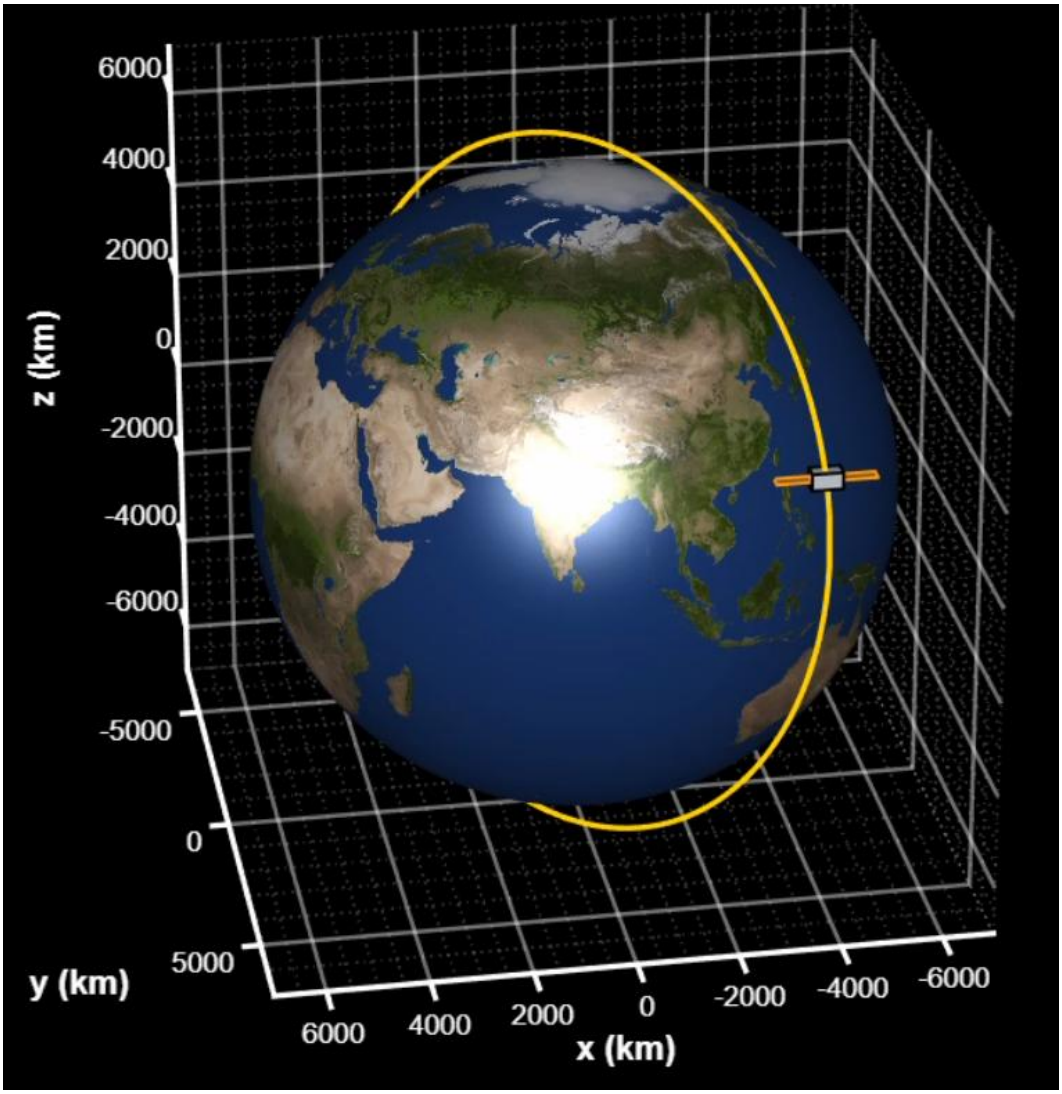

Figure 7: Visuals of a LEO Satellite Orbiting around the Earth